

ABHIYANTRIX : PILL BUDDY

PROTOTYPING AND PRODUCT DESIGN LAB

END-SEM REPORT

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CHAPTER- 1

1.1 INTRODUCTION

Medication adherence is an important factor in ensuring effective treatment and patient safety. Self-administration, irregular administration, readiness of drugs especially OTC (over the counter) and even some antibiotics are responsible for drug misuse, resistance and tolerance. To avoid these, we come to a proposed solution that includes administered drug history data, IOT based alarm/ reminders and dispense systems connected through a digital interface.

1.2 PROBLEM STATEMENT

To better understand the problem a short survey was circulated among doctors and healthcare providers. The responses indicated that missed doses, incomplete medication courses, and unmonitored self-medication are frequent problems encountered in clinical practice.

There is also a lack of reliable records of medicine consumption. Patients may self-administer medicines irregularly or combine multiple drugs without maintaining any documentation.

A patient's noncompliance to medical adherence can worsen the disease state, for instance improper maintenance of dopamine level in blood can cause paralysis during parkinsonism.

While there are pill dispensers that have a reminder system there isn't much for collecting data of over the counter and medication adherence.

Thus, there is a need for an IoT based reminder and medicine data collection system that can promote the overall treatment and lifestyle quality of the patient.

1.3 OBJECTIVES

- IOT based device that scans medicine barcodes and stores the information in a database.
- To schedule a timer to set an alarm to take the medicine.
- Dispenser that dispenses the correct medicine.
- A UI interface that allows users to monitor, schedule and manage the medication details.

1.4 BLOCK DIAGRAM

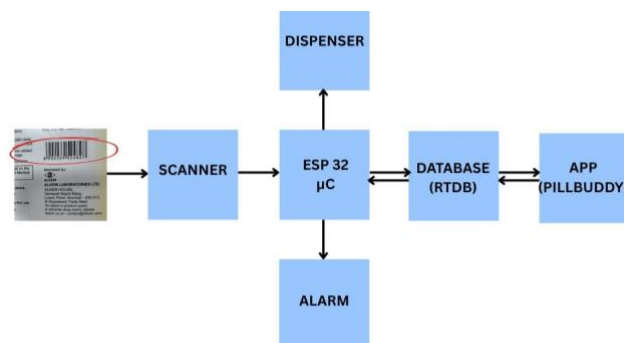


Fig. 1.1 Block Diagram

CHAPTER- 2 METHODOLOGY

2.1 SYSTEM DESIGN

2.1.1 Hardware requirements

- ESP 32
- Barcode scanner
- Buzzer
- Breadboard
- Servo motor
- PCB Board

2.1.2 Circuit design

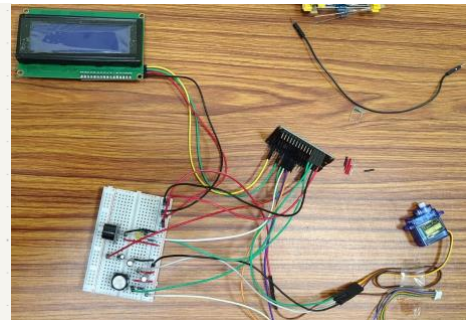
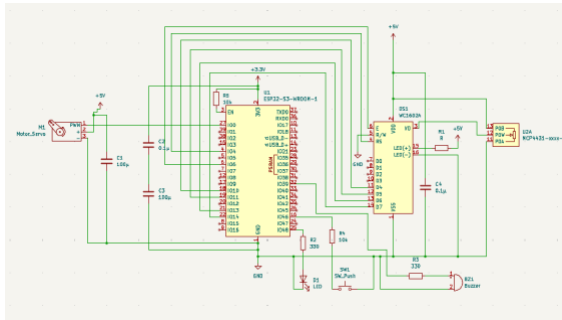


Fig. 1.2 Circuit diagram on Ki-CAD

Fig. 1.3 Hardware set up on breadboard

- Esp32 microcontroller is used for controlling the processing and system operation. It was used as it has built-in WIFI and RTC features.
- A 2D barcode scanner module with an LED indicator is used to scan the barcodes on the medicine.
- A 12C LCD Display is used to show system status and notifications.
- A buzzer, LED, push button are included for user interaction.
- A servo motor is integrated for the pill dispenser. The servo motor was coded to rotate at 30 degrees at each interval. The degree of rotation was chosen in accordance with the angle of partition of the pill dispenser.
- The circuit was initially connected and tested on a breadboard and was later transferred to a PCB.



Fig. 1.4 PCB Soldered Circuit

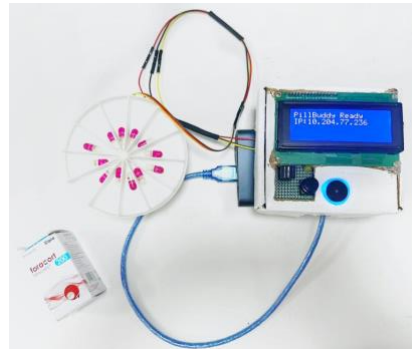
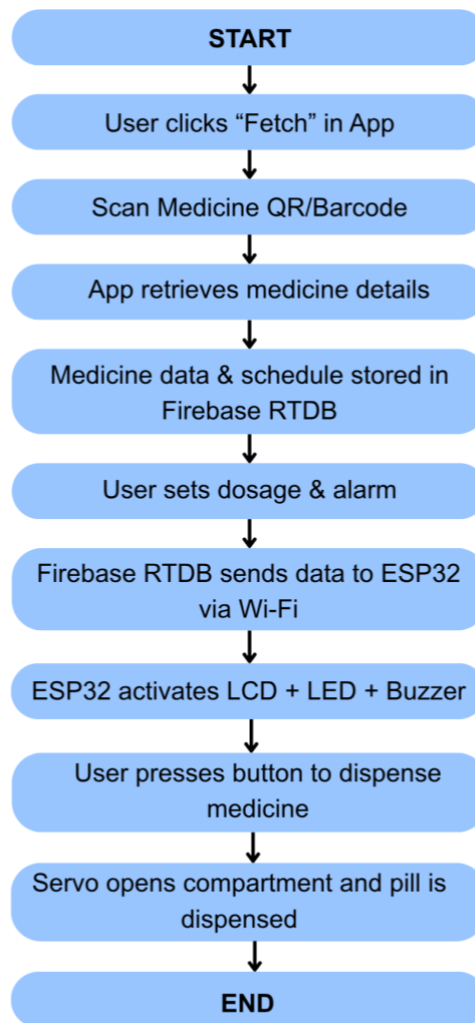


Fig. 1.5 Final Hardware set up

2.1.3 Workflow



2.2 DESIGN AND 3D PRINT

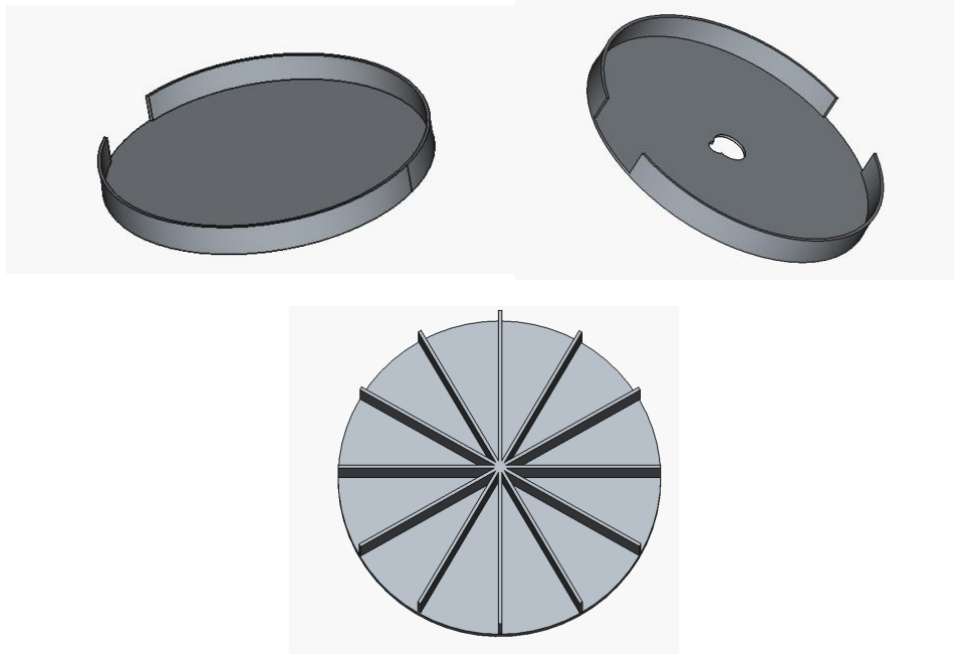


Fig. 2.1 3D design of dispenser on Free CAD

The market currently has a wide variety of dispensing systems. Our goal was to choose a technique that meets specifications such as tablet singulation with sufficient holding capacity, decreased space occupancy, and universality. The final design, which was straightforward and met every criterion, was shown above. The device casing was designed in two sections: a base and a cover. The base included a provision for motor placement, while both the base and cover incorporated openings for tablet loading and dispensing. The dispensing mechanism consisted of a cartwheel-like structure containing 12 partitions for storing tablets. Each partition was designed with a 30° separation to provide sufficient space for individual pills. The shaft of the dispensing wheel was attached to the base of the cartwheel mechanism and enclosed securely within the casing. Tablets are loaded individually, one at a time, during each cycle, and the mechanism resets its position after every rotation to enable dispensing at the user-programmed timings in synchronization with the alarm system.

2.3 APP DESIGN

A flutter-based mobile application was developed to interface with the ESP32. The application was designed to provide medicine management, reminder scheduling, barcode-based medicine identification, and synchronization with cloud databases.

Technologies used:

1. Flutter - front-end app development
2. WIFI - Seamless connection between App and ESP
3. Firebase - Login and authentication
4. Firestone - Cloud storage
5. Pub.dev- Libraries

The app provides following features:

1. Profile - It includes patient's basic info, i.e., Name, Age, Sex, Allergies.
 2. Fetch Hardware Scan - It enables the scanner to scan the medicine barcode and fetch data from firestone via RTDB.
 3. History - It saves patient medication history.
 4. Reminder - It reminds the user to take medicine at a specific time entered by the user.
- It can work in 2 ways-
- a) Local app notification on phone.
 - b) Writing data to the buzzer connected to ESP via RTDB.

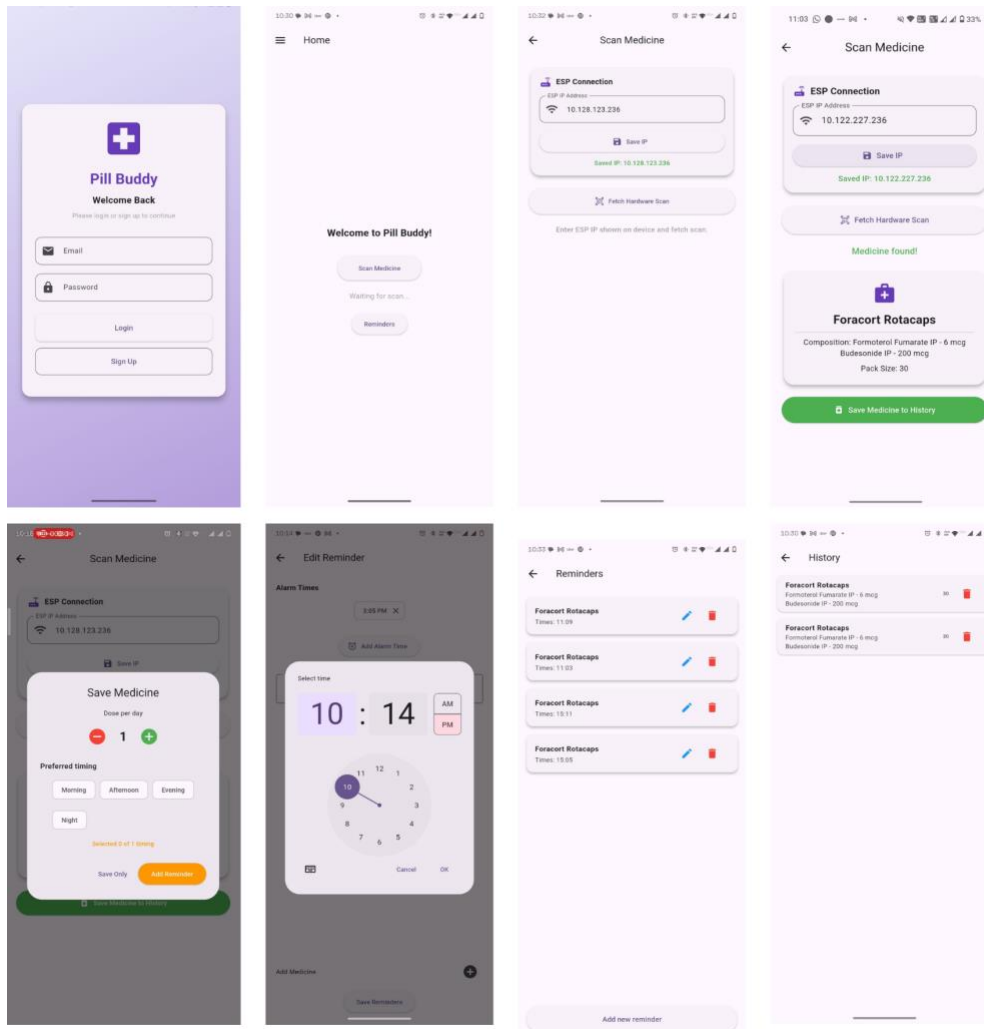


Fig. 2.2 Application UI

2.4 DATABASE

2.4.1 DATA COLLECTION

For the purpose of prototype, a pivot scale database was prepared. The data was collected from IITG Hospital pharmacy. The following details were collected:

1. Barcode
2. Proprietary Name - Brand given name
3. Composition - Active pharmaceutical Agents and Adjuvants
4. Strength - API's concentration
5. Package size – Number of tablets in given pack (bottle or blister)

2.4.2 DATABASE CREATION

Database management system (FIREBASE) is required to store Medicine Data and User data (Authentication + medicine + reminder).

The medicine data is documented with barcodes and other details in Firebase's Firestore. Firebase's cloud storage- Realtime Database (RTDB) acts as the bridge between the User and the device. It can actively read and write data from the app and the device. Thus, it can save user information, transfer the scanned data and the alarms seamlessly between the app and the microcontroller.

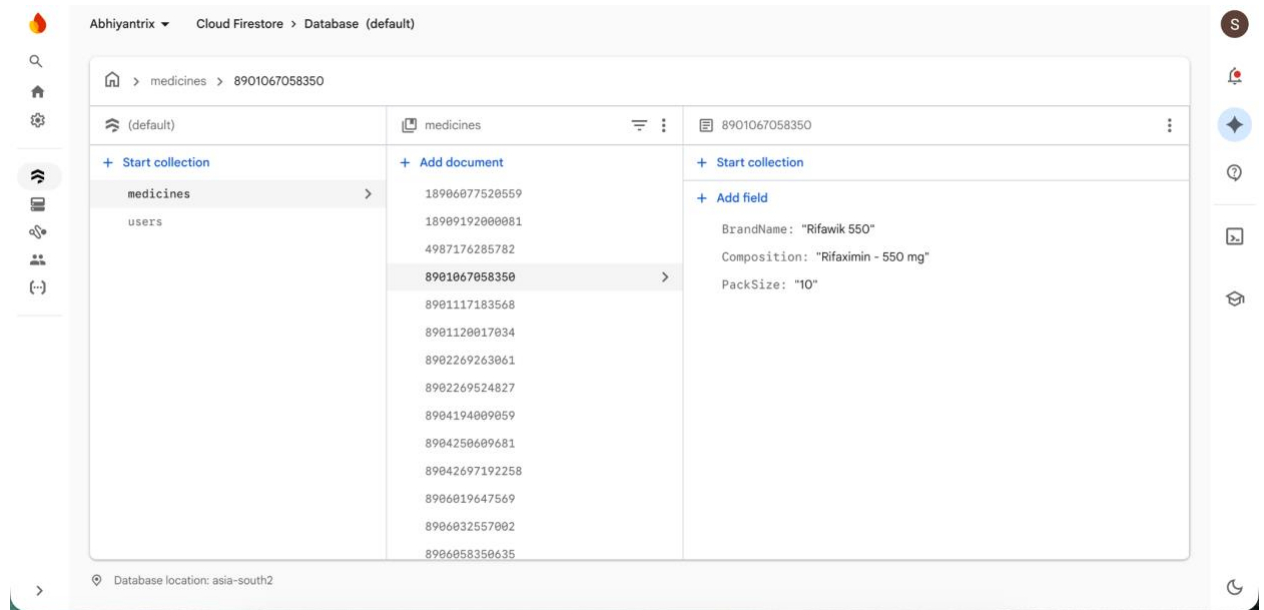


Fig. 2.3 Firestone Database

2.5 SYSTEM INTEGRATION

The proposed smart pill dispensing system integrates software, cloud database, and hardware to enable automated medicine reminders and dispensing. The app and the hardware are connected via WIFI.

A predefined medicine database is maintained in Firebase - Firestore, containing medicine information including medicine name, dosage details, usage instructions, and identification data linked to QR codes or barcodes.

When the user scans a medicine barcode/QR code through the mobile application, the application retrieves the corresponding medicine information from the Firebase database and displays it automatically.

```
{
  "rules": {
    ".read": true,
    ".write": true
  }
}
```

The user then sets the required medicine schedule and reminder timings through the application interface. These schedules are stored and synchronized in the Firebase Realtime Database, enabling real-time access across connected devices.

The ESP32 microcontroller continuously communicates with Firebase over Wi-Fi using RTDB queries to monitor reminder updates and scheduled events. At the specified time, the ESP32 activates the alert system consisting of an LED indicator and buzzer. Simultaneously, the LCD module displays reminder notifications such as “Take Medicine.”

Once the user presses the dispense button on the pillbox for confirmation, the ESP32 controls the servo motor to open the corresponding pill compartment and dispense the medicine.

CHAPTER- 3

3.1 CHALLENGES

Sr.no	Challenge	Resolution
1.	Initially, Bluetooth connectivity of the ESP module was used for communication. However, unstable connections and limited operational range affected system reliability.	To resolve this issue, the system was migrated to Wi-Fi based communication using Firebase RTDB, which provided more stable connectivity, real-time synchronization, and longer operational range.
2.	Timing synchronization issues occurred due to delays and unstable communication while using Bluetooth connectivity.	The issue was resolved by switching to Wi-Fi-based communication, which improved synchronization, responsiveness, and real-time data transfer.
3.	Incorrect GPIO pin configuration caused improper functioning of connected components and errors.	The issue was resolved by reconfiguring the GPIO mappings, verifying circuit connections, and updating the ESP32 code accordingly.
4.	Loose terminals in the initial breadboard caused contact failures and unstable circuit performance.	The issue was resolved by shifting to PCB and soldering the connections.
5.	The servo motor showed issues with the dispensing mechanism.	The issue was resolved by replacing it with a new motor along with a capacitor, which resulted in smoother dispensing and improved accuracy.
6.	Firebase RTDB initially retained previously scanned barcode data, causing outdated medicine information to be displayed in the application.	The issue was resolved by clearing the hardware-scans/latest before each new scan to ensure only the current scan data was processed.
7.	Minor misalignment between the motor and dispenser position due to calibration errors.	Despite repeated calibration and testing, slight rotational deviations still persist and remain an area for future optimization.
8.	The actual height of the motor shaft created an unforeseen gap in the dispensing mechanism, increasing the possibility of pills getting stuck.	Parts from previous 3D prints were reused and modified to close the gap and improve dispensing.

9.	The machine resolution caused minor errors between the designed dimensions and the actual printed dimensions, resulting in deviations in millimeters.	Some components were reprinted, while minor dimensional errors in other parts were managed through mechanical adjustments and workarounds.
10	Mounting and fixing components was difficult as some adhesives seeped into the gaps of the 3D-printed material, while the glue gun created uneven surfaces. A stable and flat mounting surface was required for proper motor alignment.	A makeshift mounting solution was implemented using double-sided tape combined with Fevi quick adhesive applied over the tape. This prevented adhesive seepage and protected the motor from damage while also providing a more stable mounting surface, although it slightly increased the overall height of the assembly.
11.	Third-party medicine barcode databases had licensing restrictions, and scraping data from such platforms was not a reliable or ethical solution.	Medicine information was collected from pharmacy resources and a custom local medicine database was developed for the project.
12.	Implementing QR code-based medicine identification required access to APIs or applications to retrieve medicine information, which was not feasible for the prototype. Barcode-based identification had limited applicability, as not all medicines contain barcodes.	The system was shifted to barcode-based identification for easier implementation and database integration using a custom medicine database. However, QR-based identification remains a better long-term approach due to the mandatory presence of QR codes on most medicines and its broader applicability.

3.1.1 Challenges for 3D model

Previous designs:

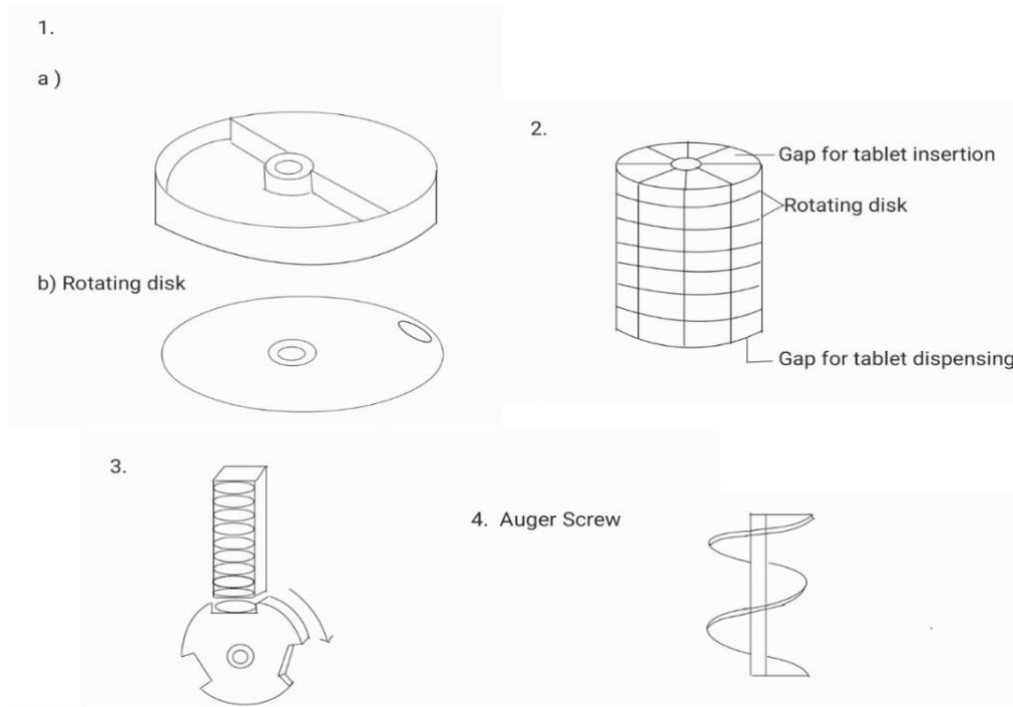


Fig. 3.1 Early-stage conceptual 3D Layout of the Pill Dispenser

These were the 4 ideas that were considered, however due to lesser feasibility and flexibility we switched to the current model.

In the Auger screw model, the pill should always be in a certain orientation for it to be picked up. Also, due to the force and friction, the pill could get cut or damaged. There were more chances of the pill getting stuck and the model malfunctioning. Due to too many complications this model had to be skipped.

In the Rotating disk model, achieving universality was a problem as different pills come in different sizes and weights. Another problem was to determine if a pill was dispensed or not, for which an IR sensor or alternative would be needed, however due to pills coming in different sizes and weights the vibrations would have to be tweaked every time we insert a new type of medicine.

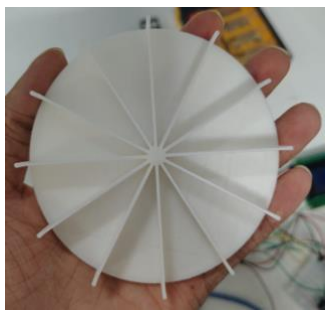


Fig. 3.2 Final 3D printed dispenser

In the current model the implementation is straightforward and simple and universality and singulation is possible. However, the volume and scale is low which is a future prospect to work on.

CHAPTER- 4

4.1 RESULTS AND DISCUSSION

- a. The proposed smart pill dispensing system was designed and integrated using hardware, cloud database services and a mobile application interface, using an IOT-based architecture the system showed that it could automate pill dispensing, keep medication records, and send scheduled medication reminders.
- b. The Firebase Realtime Database enabled synchronization between the mobile application and the ESP32 microcontroller.
- c. The hardware module used RTDB queries to retrieve medication regimens and reminder timings that were input through the application and updated in real time.
- d. The barcode/QR based medication identification system reduced the need for human entry and streamlined medication entry.
- e. Using Pharmacy resources, a prototype medicine database was made to show that tracking medications and managing data within the system is feasible.
- f. The alarm system, which included an LCD display, buzzer, and LED indication, successfully informed users when their medications were due. The pill compartment dispensing mechanism was operated by the servo motor, as confirmed by the dispense button.
- g. The produced prototype tackles several problems related to prescription non-adherence, such as inconsistent drug consumption, missing doses, and a lack of medication history recording. The suggested solutions offer extra capabilities including cloud synchronization, automated dispensing, and medication data logging in comparison to traditional reminder-based pill boxes.

4.2 FUTURE SCOPE

The future scope of this system can have some improvisation by expanding the medication database which will allow for wider usability. The system reliability on WIFI through the phone can be avoided by adding a GSM module on the device. The dispensing mechanism must be optimized to improve dispensing accuracy, speed, and overall system reliability. AI-based adherence analysis, caregiver notification systems, and interaction with hospital management systems for improved patient monitoring can also be integrated for future enhancements.

CHAPTER- 5

5.1 CONTRIBUTION OF EACH TEAM MEMBER

1) Serene Sitlhou

- Designing the circuit
- Connecting, configuring, interfacing and troubleshooting the GPIO pins with the ESP32 microcontroller and electronic components.
- Implementing and testing the circuit on a breadboard.
- Transferring the finalized circuit on PCB for improved stability, reliability and compactness
- IDE code

2) Shubham Luthra

- Front-end and the UI design of the Application.
- Configured Firebase Realtime Database (RTDB) for cloud data handling
- Connecting and optimizing the ESP connection to the Application via Bluetooth and WIFI.
- Backend to create logic for real-time data transfer.
- Optimised IDE code.

3) Priyadharshini JM

- Designed multiple 3D models for dispenser using FREECAD
- Selecting best fit model for the print.
- Optimized the design for component placement and accessibility
- Coordinated the 3D printing process for the finalized prototype design.

4) Dhanishtha Desai

- Assisted in mechanical integration of hardware components
- Planned structural arrangement for ESP32, servo motor, and dispensing unit.
- Configured integration of 3D dispenser with the electronics.
- Documentation work including Poster Design.

5) Nidhi Yadav

- Coordinated with Hospital pharmacy for Medicine database.
- Collected Data from Hospital pharmacy included with all essential information.
- Prepared local database on spreadsheets.